

6.1 Least squares as a nearest-point problem

Chapter 5 introduced least squares as a calculus problem: write down the sum of squared errors as a function of the coefficients, differentiate, set the derivative to zero. That route gets the right answer, but it hides the structure of the answer. It treats the coefficient vector $\boldsymbol{\beta}$ as the basic unknown. It needs $\mathbf{X}^\top \mathbf{X}$ to be invertible before it can say anything. And it gives no hint of why some quantities, such as fitted values, residuals and sums of squares, behave well in every model, while others, such as individual coefficients, can be fragile or not defined at all.

This chapter starts from a different place. Instead of the p -dimensional space where $\boldsymbol{\beta}$ lives, it works in the n -dimensional space where the data live.

6.1.1 Two pictures of the same data

A regression data set with n observations and a single regressor is usually drawn as a scatterplot: n points in the plane, one for each observation, with the regressor on one axis and the response on the other. Call this the **variable space** picture. It is the right picture for seeing the shape of a relationship, but it scales badly. With k regressors it needs $k + 1$ dimensions, and it has no natural place for the model's coefficients.

The other picture turns this around. Stack the n responses into a single vector $\mathbf{y} = (y_1, \dots, y_n)^\top$, and regard it as *one* point in $\mathbb{R}^n$. Each column of the model matrix $\mathbf{X}$ is also a vector in $\mathbb{R}^n$: the intercept column is $\mathbf{1}$, and the column for a regressor holds its n observed values. This is the **observation space** picture. Its dimension is the sample size, which is usually far more than three. But it contains only $p + 1$ vectors of interest, and those always span a subspace of dimension at most $p + 1$. Most of the geometry happens in a subspace small enough to draw.

In observation space, the linear model

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \boldsymbol{\beta} \in \mathbb{R}^p,$$

says one thing: the mean vector $\mathbf{E}(\mathbf{Y})$ lies in the set

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\},$$

the **column space** of $\mathbf{X}$. That set is a subspace of $\mathbb{R}^n$. The coefficients $\boldsymbol{\beta}$ are the coordinates of $\mathbf{E}(\mathbf{Y})$ with respect to the columns of $\mathbf{X}$, but the claim itself is only about *where* $\mathbf{E}(\mathbf{Y})$ is. It does not depend on which spanning vectors are used to describe the subspace.

6.1.2 The least squares principle, restated

The observed $\mathbf{y}$ is almost never in $\mathcal{C}(\mathbf{X})$, because noise moves it off the subspace. A natural estimate of $\mathbf{E}(\mathbf{Y})$ is the point of $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{y}$. With Euclidean distance, this is exactly least squares:

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2,$$

where $\mathbf{x}_{(i)}^\top$ is the i th row of $\mathbf{X}$. Minimizing over $\mathbf{b}$ is the same as searching $\mathcal{C}(\mathbf{X})$ for the point nearest to $\mathbf{y}$.

Definition 6.1.1. Least squares

A vector $\hat{\boldsymbol{\beta}} \in \mathbb{R}^p$ is a **least squares estimate** of $\boldsymbol{\beta}$ if

$$\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \|\mathbf{y} - \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the **fitted value** vector, and $\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{Y}}$ is the **residual** vector.

The definition does not assume that the minimizer is unique, and deliberately so. In [Section 6.4](#) we will see that the *nearest point* $\hat{\mathbf{Y}}$ is always unique. The coefficient vector $\hat{\boldsymbol{\beta}}$ that produces it is unique only when the columns of $\mathbf{X}$ are linearly independent.

6.1.3 A picture with three observations

The smallest case that still shows all the geometry has $n = 3$ and $p = 2$. Then $\mathcal{C}(\mathbf{X})$ is a plane through the origin in $\mathbb{R}^3$, and $\mathbf{y}$ is a point somewhere off the plane.

Example. Three observations

Take responses $\mathbf{y} = (1, 4, 2)^\top$ at regressor values $\mathbf{x} = (0, 1, 3)^\top$, with an intercept, so that $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$. The least squares fit (computed by the listing below) is

$$\hat{\mathbf{Y}} = (2.143, 2.286, 2.571)^\top, \quad \hat{\mathbf{e}} = (-1.143, 1.714, -0.571)^\top,$$

with intercept 2.143 and slope 0.143. Two facts stand out. First, the residual is orthogonal to both columns of $\mathbf{X}$: the entries of $\hat{\mathbf{e}}$ sum to zero, and $\mathbf{x}^\top \hat{\mathbf{e}} = 0$. Second, the squared lengths add:

$$\|\mathbf{y}\|^2 = 21.000 = 16.429 + 4.571 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{e}}\|^2.$$

[Figure 6.1.1](#) shows the configuration. The fitted vector is the foot of the perpendicular dropped from $\mathbf{y}$ to the plane.

```
import numpy as np

y = np.array([1.0, 4.0, 2.0])
X = np.column_stack([np.ones(3), [0.0, 1.0, 3.0]]) # intercept and one regressor

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat # the point of C(X) nearest to y
e_hat = y - y_hat # the residual vector

print("fitted ", y_hat)
print("residual", e_hat)
print("X^T e ", X.T @ e_hat) # zero: the residual is orthogonal to C(X)
```

[Run this code online](#)

The two facts in [Example \(Three observations\)](#) are not coincidences of this data set. They are the whole theory in miniature. The residual is orthogonal to the subspace *because* $\hat{\mathbf{Y}}$ is the nearest point, and the squared lengths add *because* the two pieces are orthogonal. The rest of the chapter makes these two statements precise and follows their consequences.

Idea. Fitted values are determined by the *subspace* $\mathcal{C}(\mathbf{X})$. Coefficients are determined by the *basis* used to describe it. Anything that depends only on $\mathcal{C}(\mathbf{X})$ is unchanged by reparameterization, by rescaling or recoding regressors, and by rank deficiency. This covers fitted

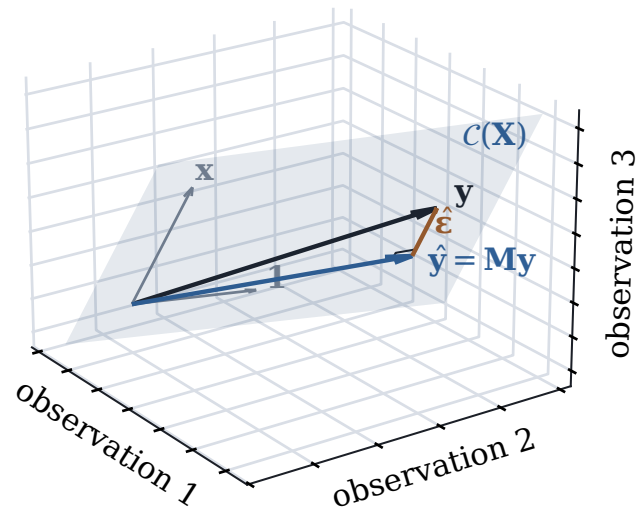


Figure 6.1.1: Least squares with $n = 3$ observations. The model's column space $\mathcal{C}(\mathbf{X})$ is the plane spanned by $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the point of the plane nearest to $\mathbf{y}$, and the residual $\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{Y}}$ is perpendicular to the plane.

values, residuals, sums of squares, R^2 and leverages.

6.1.4 Plan of the chapter

Section 6.2 collects the facts about subspaces that the geometry needs. Section 6.3 proves that nearest points exist and are unique, and identifies the matrices that compute them: the symmetric idempotent matrices, including $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ for every generalized inverse. Section 6.4 applies this to least squares. Section 6.5 studies nested subspaces, which is where sums of squares come from. Section 6.6 proves the Frisch–Waugh–Lovell theorem, which says what a single coefficient in a multiple regression measures. Section 6.7 covers reparameterization and R^2 , and Section 6.8 covers leverage. Section 6.9 replaces the Euclidean inner product with a general one, the step that leads to generalized least squares. Section 6.10 shows how the projection should be computed in floating point.

6.1.5 Exercises

6.1.5.1 A. Check your understanding

Exercise. A1

Redo Example (Three observations) in exact arithmetic. From the simple regression formulas of Theorem 5.2.1, show that the intercept is $15/7$ and the slope $1/7$, so that $\hat{\mathbf{Y}} = (15, 16, 18)^\top / 7$ and $\hat{\mathbf{e}} = (-8, 12, -4)^\top / 7$. Check exactly that $\mathbf{1}^\top \hat{\mathbf{e}} = 0$, $\mathbf{x}^\top \hat{\mathbf{e}} = 0$ and $\|\hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{e}}\|^2 = \|\mathbf{y}\|^2 = 21$.

Solution. Here $\sum x_i = 4$, $\sum x_i^2 = 10$, $\sum y_i = 7$ and $\sum x_i y_i = 10$, so $S_{xx} = 10 - 4^2/3 = 14/3$ and $S_{xy} = 10 - 4 \cdot 7/3 = 2/3$. The slope is $S_{xy}/S_{xx} = 1/7$ and the intercept $\bar{y} - \bar{x}/7 = 7/3 - 4/21 = 15/7$. Evaluating $15/7 + x_i/7$ at $x = 0, 1, 3$ gives $\hat{\mathbf{Y}}$, and $\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{Y}}$ as stated. Then $\mathbf{1}^\top \hat{\mathbf{e}} = (-8 + 12 - 4)/7 = 0$ and $\mathbf{x}^\top \hat{\mathbf{e}} = (0 \cdot (-8) + 1 \cdot 12 + 3 \cdot (-4))/7 = 0$. Finally $\|\hat{\mathbf{Y}}\|^2 = (225 + 256 + 324)/49 = 805/49$ and $\|\hat{\mathbf{e}}\|^2 = (64 + 144 + 16)/49 = 224/49$, which sum to $1029/49 = 21 = 1 + 16 + 4$.

Exercise. A2

Let $\mathbf{y} = (1, 3)^\top$ and let $\mathbf{X}$ be the 2×2 matrix whose two columns both equal $\mathbf{1}$. Show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ depends on $\mathbf{b}$ only through $b_1 + b_2$, and that the least squares estimates of Definition 6.1.1 are exactly the $\mathbf{b}$ with $b_1 + b_2 = 2$. Which of $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{Y}}$, $\hat{\mathbf{e}}$ does the definition determine uniquely here?

Solution. $\mathbf{X}\mathbf{b} = (b_1 + b_2)\mathbf{1}$, so with $t = b_1 + b_2$ the criterion is $(1 - t)^2 + (3 - t)^2 = 2(t - 2)^2 + 2$, minimized exactly at $t = 2$. The estimates therefore form the line $\{\mathbf{b} : b_1 + b_2 = 2\}$, while $\hat{\mathbf{Y}} = (2, 2)^\top$ and $\hat{\mathbf{e}} = (-1, 1)^\top$ are the same for every one of them. The fitted vector and the residual are determined; the coefficient vector is not.

Exercise. A3

Let $\hat{\mathbf{Y}}$ be a nearest point to $\mathbf{y}$ in $\mathcal{C}(\mathbf{X})$. Show directly from Definition 6.1.1, with no formula for $\hat{\mathbf{Y}}$, that $c\hat{\mathbf{Y}}$ is a nearest point to $c\mathbf{y}$ for every $c \in \mathbb{R}$, and that $\hat{\mathbf{Y}} + \mathbf{u}$ is a nearest point to $\mathbf{y} + \mathbf{u}$ for every $\mathbf{u} \in \mathcal{C}(\mathbf{X})$. Deduce that adding $\mathbf{u}$ leaves the residual unchanged.

6.1.5.2 B. Practice

Exercise. B1

The nearest point is unique, and this can be proved before any of the machinery of Section 6.3. Let $\mathbf{u}, \mathbf{v} \in \mathcal{C}(\mathbf{X})$ both attain $d = \min_{\mathbf{b}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|$. Expand inner products, or use the parallelogram law of Exercise (A2), to verify

$$\left\| \mathbf{y} - \frac{\mathbf{u} + \mathbf{v}}{2} \right\|^2 = \frac{1}{2} \|\mathbf{y} - \mathbf{u}\|^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{v}\|^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2,$$

and conclude that $\mathbf{u} = \mathbf{v}$. Where does the argument use that $\mathcal{C}(\mathbf{X})$ contains the midpoint of any two of its elements?

Solution. Write $\mathbf{y} - (\mathbf{u} + \mathbf{v})/2 = \frac{1}{2}(\mathbf{y} - \mathbf{u}) + \frac{1}{2}(\mathbf{y} - \mathbf{v})$ and expand:

$$\left\| \mathbf{y} - \frac{\mathbf{u} + \mathbf{v}}{2} \right\|^2 = \frac{1}{4} \|\mathbf{y} - \mathbf{u}\|^2 + \frac{1}{4} \|\mathbf{y} - \mathbf{v}\|^2 + \frac{1}{2} (\mathbf{y} - \mathbf{u})^\top (\mathbf{y} - \mathbf{v}).$$

Also $\|\mathbf{u} - \mathbf{v}\|^2 = \|(\mathbf{y} - \mathbf{v}) - (\mathbf{y} - \mathbf{u})\|^2 = \|\mathbf{y} - \mathbf{u}\|^2 + \|\mathbf{y} - \mathbf{v}\|^2 - 2(\mathbf{y} - \mathbf{u})^\top (\mathbf{y} - \mathbf{v})$. Eliminating the cross term between the two displays gives the identity. Now $\|\mathbf{y} - \mathbf{u}\| = \|\mathbf{y} - \mathbf{v}\| = d$, so the left side is $d^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2$. The midpoint lies in $\mathcal{C}(\mathbf{X})$, so the left side is at least d^2 by the

definition of d ; hence $\|\mathbf{u} - \mathbf{v}\| = 0$. Without the midpoint the inequality $\|\mathbf{y} - (\mathbf{u} + \mathbf{v})/2\| \geq d$ would not be available.

Exercise. B2

Adding a constant to every response is the commonest change of units, and only a model with an intercept handles it in the obvious way.

- Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Using [Exercise \(A3\)](#), show that the fit for $\mathbf{y} + c\mathbf{1}$ is $\hat{\mathbf{Y}} + c\mathbf{1}$ and that the residual is unchanged.
- Take $n = 2$, $\mathbf{X} = \mathbf{x} = (1, 2)^\top$ and no intercept. Show that the fit for $\mathbf{y} = (1, 1)^\top$ is $(3/5, 6/5)^\top$, and that the fit for $\mathbf{y} + \mathbf{1} = (2, 2)^\top$ is $(6/5, 12/5)^\top$, not $(8/5, 11/5)^\top$. Which residual changes, and why is there no contradiction with (a)?

Solution.

- When $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, the map $\mathbf{v} \mapsto \mathbf{v} + c\mathbf{1}$ is a bijection of $\mathcal{C}(\mathbf{X})$ onto itself, and $\|(\mathbf{y} + c\mathbf{1}) - (\mathbf{v} + c\mathbf{1})\| = \|\mathbf{y} - \mathbf{v}\|$, so the distances to be minimized match up in pairs. The nearest point to $\mathbf{y} + c\mathbf{1}$ is therefore $\hat{\mathbf{Y}} + c\mathbf{1}$, and subtracting leaves $\hat{\mathbf{e}}$ unchanged. (This is [Exercise \(A3\)](#) with $\mathbf{u} = c\mathbf{1}$.)
- Minimizing $(1 - b)^2 + (1 - 2b)^2$ gives $10b = 6$, $b = 3/5$, and $b\mathbf{x} = (3/5, 6/5)^\top$. For $\mathbf{y} = (2, 2)^\top$, $10b = 12$, $b = 6/5$, and $b\mathbf{x} = (6/5, 12/5)^\top$. The residual changes from $(2/5, -1/5)^\top$ to $(4/5, -2/5)^\top$. There is no contradiction: $\mathbf{1} \notin \mathcal{C}(\mathbf{x})$, so $\mathbf{y} + \mathbf{1}$ is not $\mathbf{y}$ plus an element of the model space, and (a) does not apply. The intercept column is what makes a shift of the response a movement inside the model space.

Exercise. B3

Let $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X}_2)$, and let $d_k = \min\{\|\mathbf{y} - \mathbf{v}\| : \mathbf{v} \in \mathcal{C}(\mathbf{X}_k)\}$.

- Show $d_2 \leq d_1$, with equality iff some nearest point of $\mathbf{y}$ in $\mathcal{C}(\mathbf{X}_1)$ is also a nearest point in $\mathcal{C}(\mathbf{X}_2)$. Conclude that appending a column to $\mathbf{X}$ can never raise the residual sum of squares.
- Equality does not require the new column to be redundant. With $n = 3$, $\mathbf{X} = \mathbf{1}$ and $\mathbf{y} = (1, 2, 3)^\top$, the fit is $\hat{\mathbf{Y}} = (2, 2, 2)^\top$ and $\hat{\mathbf{e}} = (-1, 0, 1)^\top$. Show that appending $\mathbf{z} = (1, -2, 1)^\top$ enlarges the column space but leaves $\hat{\mathbf{Y}}$ and d unchanged.

6.1.5.3 C. Going deeper

Exercise. C1

[Definition 6.1.1](#) takes for granted that a minimizer exists. Prove it, without orthonormal bases.

- Reduce to the case in which the columns of $\mathbf{X}$ are linearly independent.
- Show that $c = \min\{\|\mathbf{X}\mathbf{b}\| : \|\mathbf{b}\| = 1\}$ is positive, so $\|\mathbf{X}\mathbf{b}\| \geq c\|\mathbf{b}\|$ for every $\mathbf{b}$.
- Deduce that the infimum of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ over $\mathbb{R}^p$ equals its infimum over a closed ball, and that it is attained.

- (d) Show that the conclusion can fail for sets that are not subspaces: give a nonempty $\mathcal{S} \subseteq \mathbb{R}^2$ and a $\mathbf{y}$ with no nearest point in $\mathcal{S}$.

Solution.

- (a) Discard columns that are linear combinations of the others until the rest are independent. This changes neither $\mathcal{C}(\mathbf{X})$ nor the set of attainable values $\mathbf{X}\mathbf{b}$, so it changes neither the infimum nor whether it is attained.
- (b) The function $\mathbf{b} \mapsto \|\mathbf{X}\mathbf{b}\|$ is continuous and the unit sphere of $\mathbb{R}^p$ is closed and bounded, so the minimum c is attained at some $\mathbf{b}_0$ with $\|\mathbf{b}_0\| = 1$. If $c = 0$ then $\mathbf{X}\mathbf{b}_0 = \mathbf{0}$ with $\mathbf{b}_0 \neq \mathbf{0}$, contradicting independence. Homogeneity gives $\|\mathbf{X}\mathbf{b}\| \geq c\|\mathbf{b}\|$ for all $\mathbf{b}$.
- (c) At $\mathbf{b} = \mathbf{0}$ the criterion is $\|\mathbf{y}\|$. If $\|\mathbf{b}\| > 2\|\mathbf{y}\|/c$, then $\|\mathbf{y} - \mathbf{X}\mathbf{b}\| \geq \|\mathbf{X}\mathbf{b}\| - \|\mathbf{y}\| \geq c\|\mathbf{b}\| - \|\mathbf{y}\| > \|\mathbf{y}\|$. So the infimum over $\mathbb{R}^p$ equals the infimum over the closed ball $\|\mathbf{b}\| \leq 2\|\mathbf{y}\|/c$, which is closed and bounded; a continuous function attains its minimum there.
- (d) Let $\mathcal{S} = \{\mathbf{v} : \|\mathbf{v}\| < 1\}$ and $\mathbf{y} = (2, 0)^\top$. Distances from $\mathbf{y}$ to $\mathcal{S}$ have infimum 1, approached along $\mathbf{v} = (t, 0)^\top$ as $t \uparrow 1$, but no point of $\mathcal{S}$ is at distance 1. What fails is that $\mathcal{S}$ is not closed; it is both convex and bounded.

Exercise. C2

Write $\mathcal{N}(\mathbf{X}) = \{\mathbf{d} \in \mathbb{R}^p : \mathbf{X}\mathbf{d} = \mathbf{0}\}$, and let $\hat{\boldsymbol{\beta}}_0$ be any least squares estimate.

- (a) Using [Exercise \(B1\)](#), show that the set of least squares estimates is exactly $\hat{\boldsymbol{\beta}}_0 + \mathcal{N}(\mathbf{X})$.
- (b) Deduce that the estimate is unique iff $\mathbf{X}$ has full column rank, and that otherwise the solution set is a translate of a subspace of dimension $p - \text{rank}(\mathbf{X})$ (Theorem 1.2.2).
- (c) Show that for $\mathbf{c} \in \mathbb{R}^p$ the number $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ is the same for every least squares estimate iff $\mathbf{c}^\top \mathbf{d} = 0$ for every $\mathbf{d} \in \mathcal{N}(\mathbf{X})$.
- (d) Which $\mathbf{c}$ qualify in [Exercise \(A2\)](#)?

[Chapter 8](#) takes the condition in (c) as the definition of an estimable function.

Solution.

- (a) By [Exercise \(B1\)](#) there is a single vector $\hat{\mathbf{Y}}$ attaining the minimum, so $\mathbf{b}$ is a least squares estimate iff $\mathbf{X}\mathbf{b} = \hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}_0$, that is, iff $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}_0) = \mathbf{0}$.
- (b) The set is a single point iff $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, which is full column rank; its dimension is $p - \text{rank}(\mathbf{X})$ by the rank-nullity theorem.
- (c) Every estimate is $\hat{\boldsymbol{\beta}}_0 + \mathbf{d}$, so the values taken by $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ are $\mathbf{c}^\top \hat{\boldsymbol{\beta}}_0 + \mathbf{c}^\top \mathbf{d}$ as $\mathbf{d}$ ranges over $\mathcal{N}(\mathbf{X})$. These agree for all $\mathbf{d}$ iff $\mathbf{c}^\top \mathbf{d} = 0$ throughout $\mathcal{N}(\mathbf{X})$; if $\mathbf{c}^\top \mathbf{d} \neq 0$ for some $\mathbf{d}$, the two estimates $\hat{\boldsymbol{\beta}}_0$ and $\hat{\boldsymbol{\beta}}_0 + \mathbf{d}$ give different values.
- (d) There $\mathcal{N}(\mathbf{X})$ is spanned by $(1, -1)^\top$, so the condition is $c_1 = c_2$: the sum $\beta_1 + \beta_2$ is determined by the data, and no other linear function is, apart from its multiples.